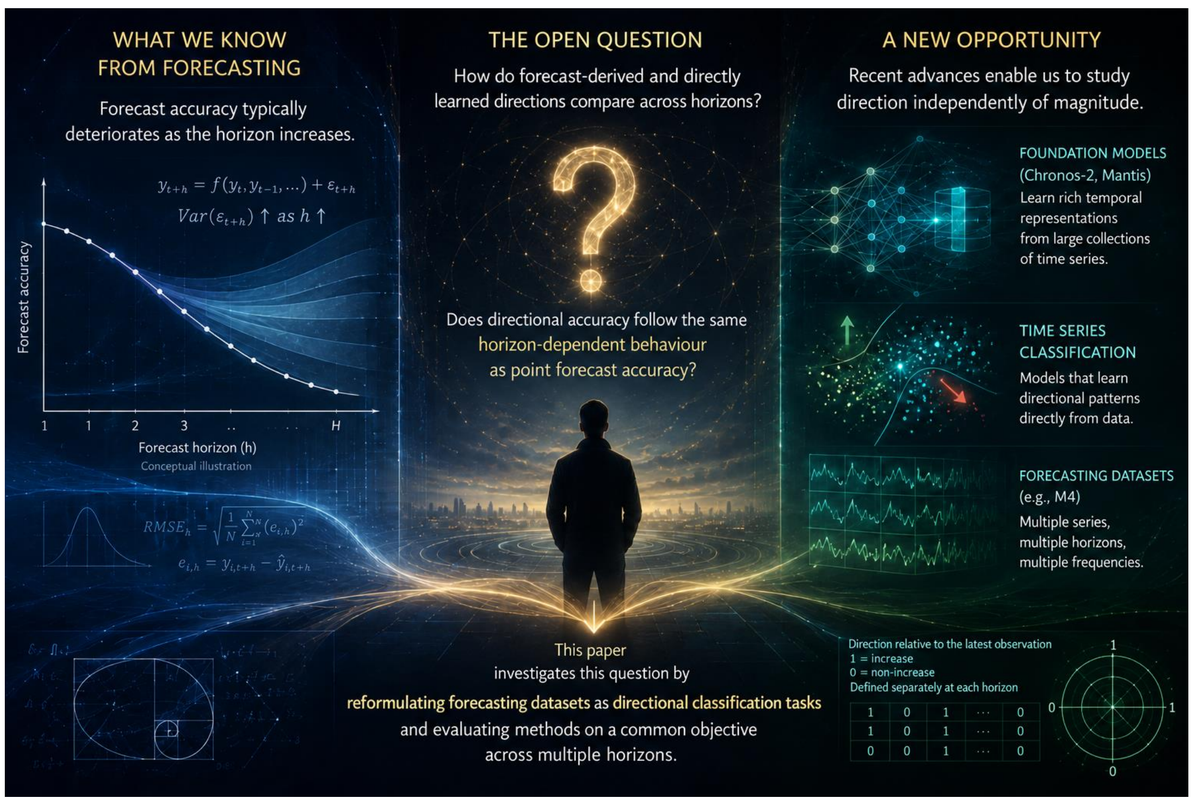
A Step in the Right Direction

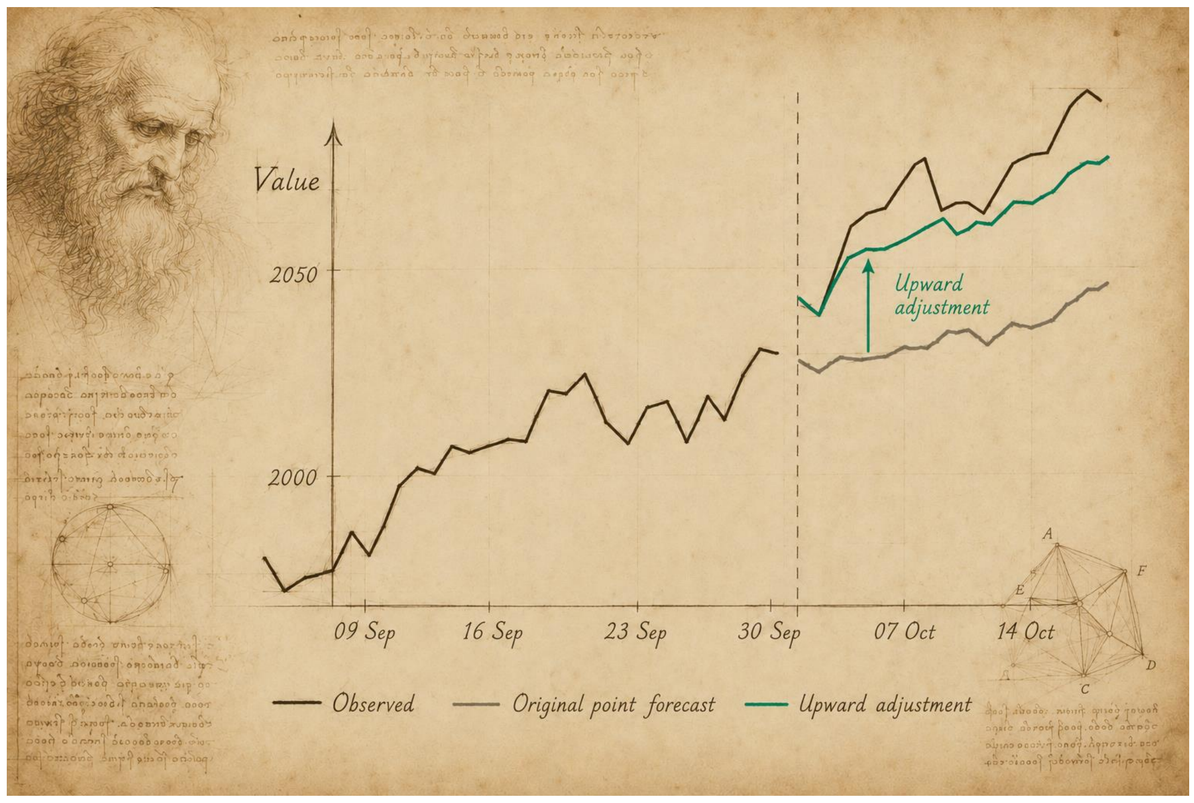
Bridging Forecasting and Classification for Temporal Foundation Models

Rafael Montano Pablo Montero-Manso Andrey Vasnev

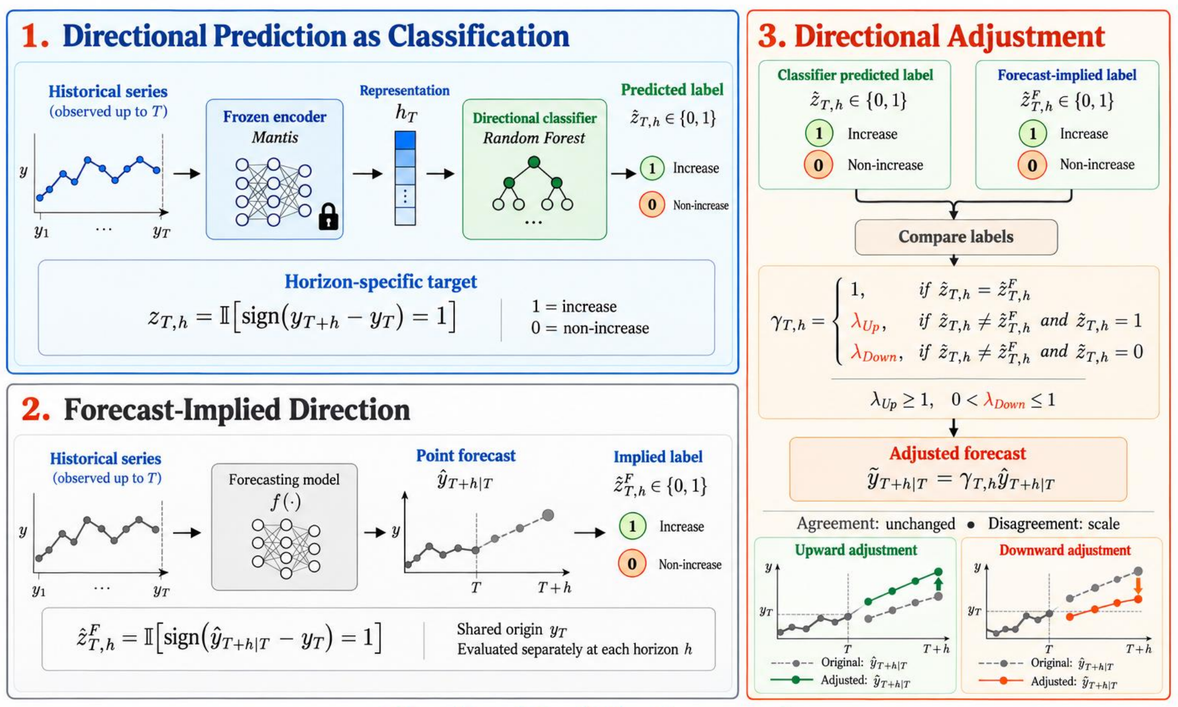
FMTS workshop, NeurIPS 2026, Sydney

Do Forecasts and Classifiers Agree on Direction?

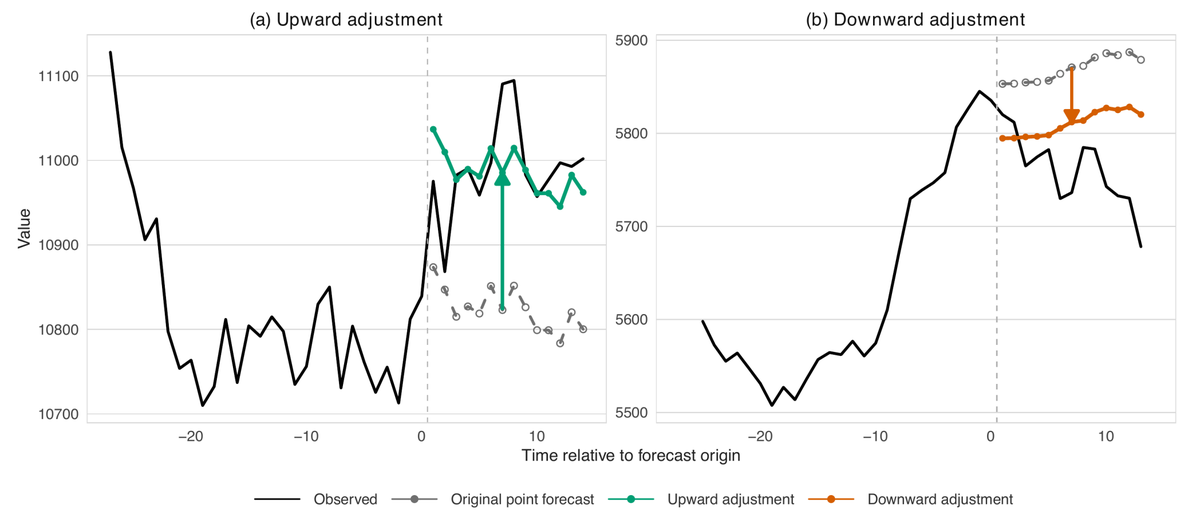




Directional Prediction Framework



Directional Prediction Framework
Compare learned and forecast-implied directions across horizons; adjust on disagreement.

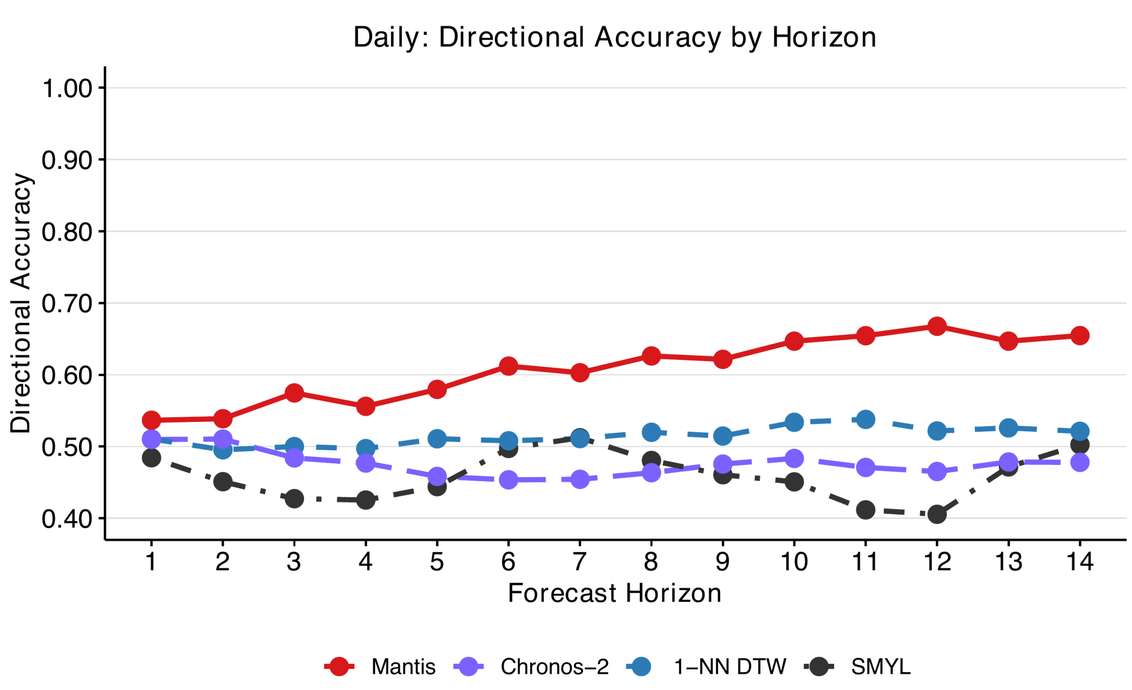


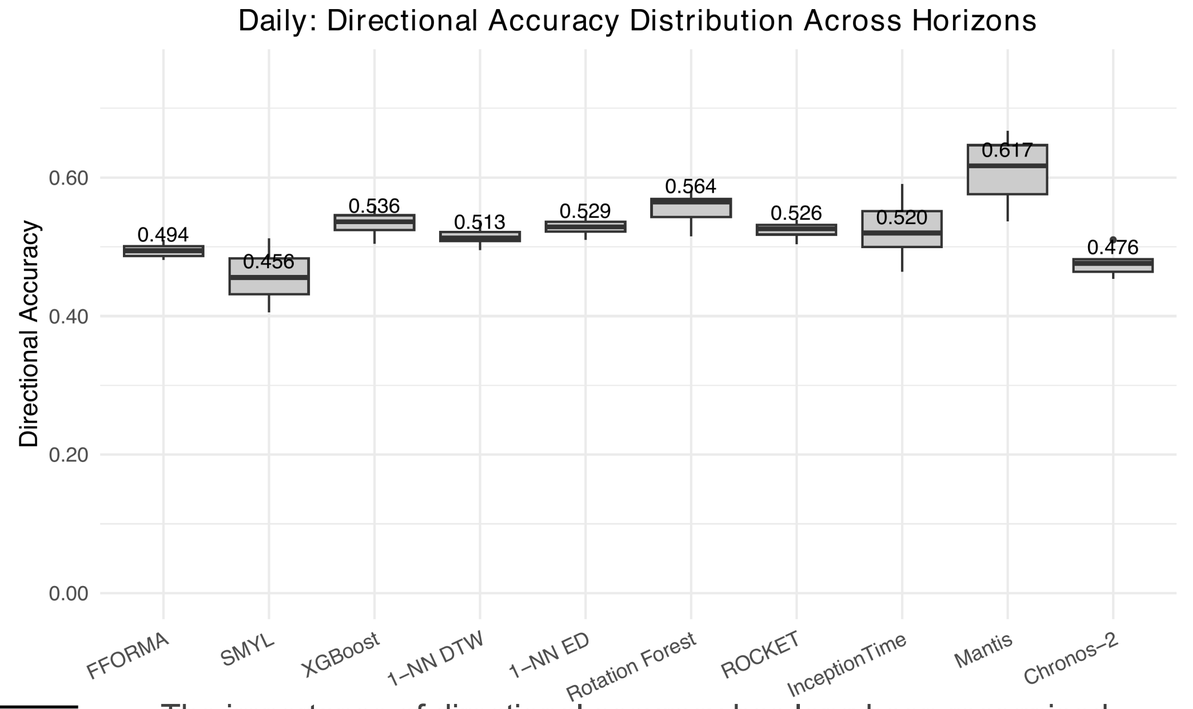
Left: M4 Daily D1602 — SMYL–Mantis upward adjustment.
Right: M4 Weekly W39 — Chronos-2–Mantis downward adjustment.

- One directional task connects forecasting and classification.
- Classifier signals guide adjustments on disagreement.

What Can Directional Signals Add to Forecasting?

M4 Forecasting Dataset · Daily (n = 4,227; H = 14)
Plots: DA across horizons · Table: DA at h = 14; sMAPE, MASE and OWA over all 14 horizons.*





Method	sMAPE↓	MASE↓	DA↑	OWA↓	Improvement over SMYL (%)		
					sMAPE↑	MASE↑	OWA↑
Chronos-2	3.09	3.38	0.478	1.023	2.61	1.88	2.24
FFORMA	3.10	3.34	0.510	1.019	2.29	2.95	2.62
SMYL	3.17	3.45	0.502	1.046	0.00	0.00	0.00
SMYL–Mantis	3.05	3.30	**0.670**	1.004	3.69	4.33	4.01
Chronos-2–Mantis	**2.95**	**3.16**	0.665	**0.966**	**6.86**	**8.41**	**7.64**
SMYL–Oracle**	1.72	1.76	0.859	0.551	45.64	49.02	47.34

* Mantis-guided OWA gains are in sample; held-out performance untested.
** SMYL–Oracle is an infeasible ex post benchmark.

- The importance of directional accuracy has long been recognised (Leitch and Tanner, 1991; Granger and Pesaran, 2000).
- Yet directional prediction has traditionally been treated as a secondary outcome of point forecasting.

- **A Step in the Right Direction — Our Contributions:**
  - A shared framework connecting forecasting and classification.
  - Evidence of contrasting directional behaviour across methods, frequencies, and horizons.
  - Exploratory forecasting value of classifier-derived directional signals.